

Microcontrollers

1. Microcontroller (MCU) – Compact IC with CPU, memory, and I/O.
2. CPU – Core unit that executes instructions.
3. Program Memory (Flash) – Stores the code permanently.
4. RAM – Temporary memory for data during execution.
5. EEPROM – Non-volatile memory for small data storage.
6. Clock – Provides timing for operations.
7. Oscillator – Generates clock signal.
8. GPIO – Pins used for input/output operations.
9. Port – Group of GPIO pins.
10. Register – Small storage inside CPU for quick access.
11. Instruction Set – Collection of commands MCU can execute.
12. Interrupt – Signal that temporarily halts main program.
13. ISR (Interrupt Service Routine) – Function executed on interrupt.
14. Timer – Counts clock pulses for delays/timing.
15. Counter – Counts external events.
16. PWM (Pulse Width Modulation) – Controls analog output using digital signals.
17. ADC (Analog to Digital Converter) – Converts analog signal to digital.
18. DAC (Digital to Analog Converter) – Converts digital signal to analog.
19. UART – Serial communication protocol (asynchronous).
20. SPI – High-speed synchronous communication protocol.
21. I2C – Two-wire communication protocol.
22. Watchdog Timer – Resets system if it hangs.
23. Bootloader – Program that loads firmware at startup.
24. Firmware – Software programmed into MCU.
25. Embedded System – System with dedicated function using MCU.
26. Power Consumption – Energy used by MCU.
27. Sleep Mode – Low-power operating state.
28. Bus – Path for data transfer.
29. Stack – Memory for function calls and returns.
30. Polling – Continuous checking of device status.